*Russian Geology and Geophysics*
*Vol. 46, No. 11, pp. 1107-1120, 2005*

*Geologiya i Geofizika*
*UDC 550.384,550.93,551.242.1*

# PALEOMAGNETISM OF THE PERMIAN-TRIASSIC TRAPS FROM THE KUZNETSK BASIN, SOUTHERN SIBERIA

A.Yu. Kazansky, D.V. Metelkin, V.Yu. Bragin, and L.V. Kungurtsev

*Institute of Geology, Siberian Branch of the RAS, 3 prosp. Akad. Koptyuga, Novosibirsk, 630090, Russia*

**In this paper we generalize the data obtained for the Kuznetsk Basin traps studied over many years. A rich collection of samples from the Permian and Triassic igneous/sedimentary succession of the Kuznetsk Basin was analyzed. As a result, directions of characteristic remanent magnetization have been established ($D_s$ = 227.1, $I_s$ = –64.3, $\alpha_{95}$ = 2.8). Also, the paleomagnetic pole has been located accurately enough to be used in paleotectonic reconstructions. The Permian-Triassic age of the characteristic component of magnetization has been confirmed by fold and baked-contact tests. Comparative analysis of the magnetic zoning record in the sections of traps of the Kuznetsk Basin, northern Siberian Platform and West Siberian Plate shows that the processes of intraplate magmatism were mutually related in these areas about the time of the Permian-Triassic transition. Systematization of the obtained and available paleomagnetic data shows that the trap magmatism in the Kuznetsk Basin, as compared with the rest of Siberia, was an extremely short-term event. The entire section corresponds to the initial stage of the intraplate magmatic activity in the Siberian region, with their formation lasting no more than 1 Myr. The position of the calculated paleomagnetic pole (PLat = 60.0, PLong = 172.7, $\alpha_{95}$ = 4.0) and the orientation of the Permo-Triassic latitudes of the Kuznetsk Basin have a considerable angular deviation from those of the Siberian Platform (PLat = 49.0, PLong = 151.6, $\alpha_{95}$ = 5.0). The angular distance makes about 24° of the large circle and is due to intraplate strike-slip displacements. The nature of the strike-slip movements is interpreted in terms of plume tectonics and is governed by the clockwise rotation of the Siberian part of the Eurasian continent relative to the European part. Thus, the formation of the Permian-Triassic volcanotectonic structures of the Kuznetsk Basin followed the "pull-apart" scheme.**

***Trap formation, Permian-Triassic, paleomagnetism, intraplate displacement, Kuznetsk Basin***

## INTRODUCTION

The Permian-Triassic boundary in Asia is marked by the immense manifestation of trap magmatism in the north and granitoid alkaline bimodal magmatism within the Ural-Mongolo-Okhotsk foldbelt. According to Dobretsov [1], these events are linked to the activity of a system of coeval or nearly coeval mantle plumes grouped into a Permian-Triassic superplume. The Siberian trap formation includes traps of the Siberian Platform, Taimyr, Kuznetsk Basin, and buried covers of the West Siberian Plate (Fig. 1) [1–4], The trap formation discordantly overlaps the Paleozoic and Proterozoic complexes, the youngest of which are sedimentary rocks of Permian age. The sedimentary interbeds within the trap formation, which are rather sparse, contain both Permian and Triassic complexes of flora [5], which suggests that the Permian-Triassic boundary runs inside the trap formation [6]. Isotope-geochronological studies provide a narrow time range of formation of the Siberian traps. In the Tunguska basin, the Ar-Ar ages of the Morongo and Mokulaev Formations average 244 Ma [7]. Biotite and plagioclase Ar/Ar ages of the Noril'sk-1 and Talnakh intrusions are 249±2 Ma [8]. For the rocks of the Ivakino, Gudchikha, and Nerakar Formations (Noril'sk district and Putorana Plateau) Ar-Ar ages average 248.8±2.4 Ma [9]. Zircon U-Pb age of the Noril'sk intrusion is estimated at 248.0±3.7 Ma [10]. Close ages were also inferred from volcanites penetrated by superdeep BH-19 in the Noril'sk district (245.5±4.5 and 248.4±3.7) [11]. Obviously, the obtained

©2005 UIGGM, SIBERIAN BRANCH OF THE RAS

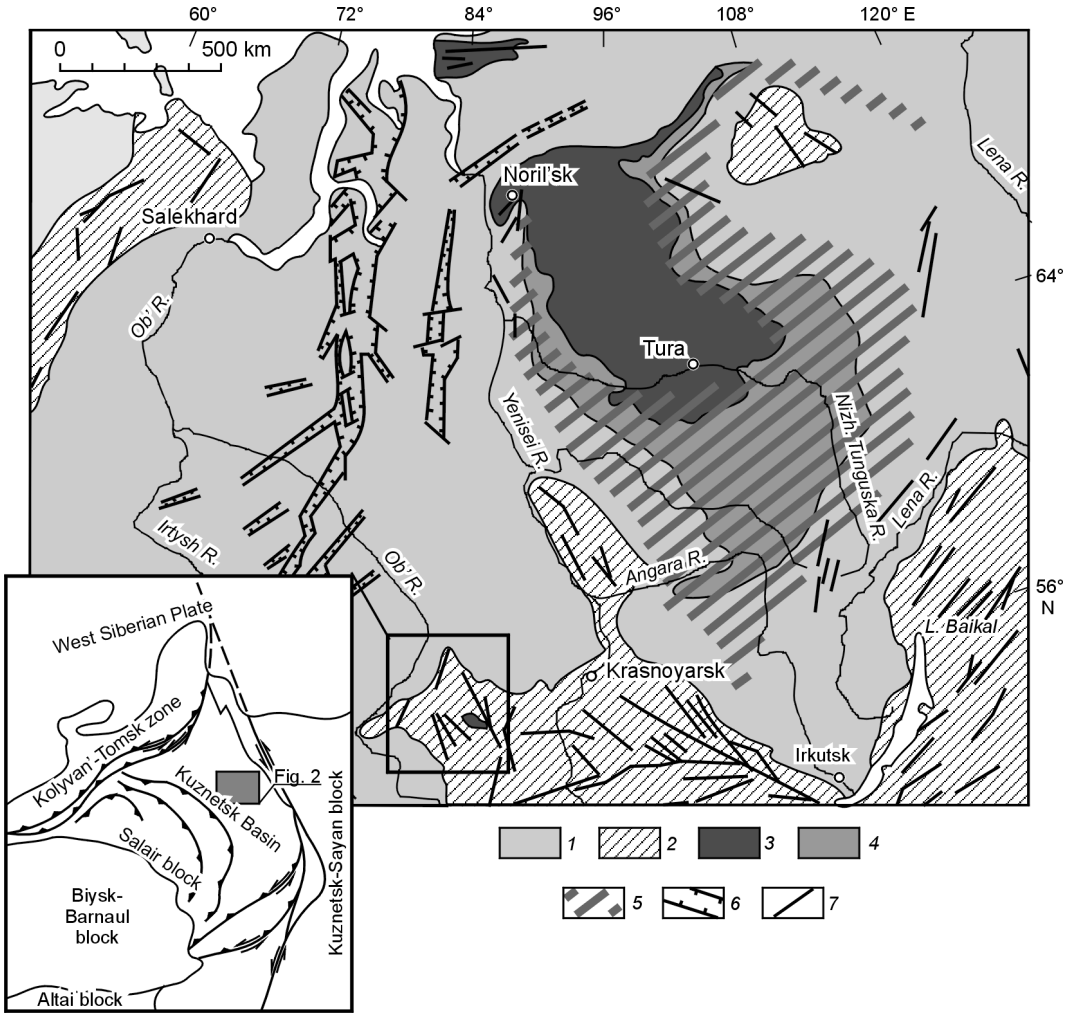



**Fig. 1. Schematic distribution of Permian-Triassic traps of Siberia (after [2], supplemented and simplified). The inset shows basic structural units of the study area [43].** ***1*** **— sedimentary cover of the Phanerozoic;** ***2*** **— foldbelts and inliers of the Precambrian basement;** ***3–5*** **— trap formation:** ***3*** **— basalts,** ***4*** **— basalt tuffs and tuffites,** ***5*** **— areas of occurrence of intrusive traps;** ***6*** **— rift structures of the pre-Jurassic basement of the West Siberian Plate;** ***7*** **— main tectonic disturbances.**

data show a good convergence and are very close to the age estimates obtained for the Permian-Triassic boundary, 248–251 Ma [12]. Abundance of the trap rocks and a relatively narrow interval of their formation permit the use of the Permian-Triassic episode of magmatic activity as an interregional reference for correlation of events on the Siberian Platform and adjacent territories. Paleomagnetic studies of geological complexes corresponding to this large-scale event permit us to reconstruct the intraplate movements within the Eurasian continent, as it was made, for example, in [13]. The most promising object for answering the question about the extent of the Late Paleozoic-Mesozoic strike-slip motions in the western part of the Altai-Sayan area is the trap formation of the Kuznetsk Basin, where results of the reconnaissance paleomagnetic studies performed in the early 1970s [14] suggest that the motions of this kind actually took place.

## SHORT GEOLOGICAL CHARACTERISTIC OF OBJECTS OF PALEOMAGNETIC STUDIES

The trap formation of the Kuznetsk Basin is represented by the Mal'tsev Group, whose rocks form three areas in the eastern part of the trough. They discordantly lie upon the Upper Permian coaliferous deposits and are,

also discordantly, overlain by the Jurassic sedimentary complexes [3]. The bottom of the group is composed of greenish-gray siltstones with interbeds of sandstones, which grade upsection into tuffaceous siltstones, tuffaceous sandstones, tuffs with thick tabular bodies of basalts and microdolerites. The upper beds are gray siltstones and sandstones with an interbed of conglomerates in the middle part and subtabular bodies of porphyraceous basalts and microdolerites at the top. The igneous rocks of the group represented by the cover and subvolcanic phases are distinguished as the Abinka trap trachybasalt complex [15]. The amount of volcanogenic material increases northward and within the Saltymakovsky Ridge; in the north of the Bungarap syncline (Fig. 2), the section is dominated by basalts and subvolcanic intrusions [3]. The group is crowned by a thick horizon of conglomerates. The complex of asterozoanes and flora dates the deposits as the Early Triassic and the transition to the Middle Triassic [16]. According to Kutolin [3], the time of formation of basalts of the trap formation is the Early Triassic.

Samples for paleomagnetic studies were taken from four sites of the section of the Mal'tsev Group within two basic volcanotectonic structures of the Astashkin brachysyncline in the south of the region and in the Bungarap syncline in the north (see Fig. 2).

**Site I** lies in the spurs of the Karakan Ridge in the southwest of the Bungarap syncline (54.42° N, 86.82° E). Basalts of four successive subtabular bodies were sampled there in a quarry at the bottom of trap formation of the Kuznetsk Basin and underlying sandstones of the Upper Permian. The contact between basalts and sandstones is not exposed.

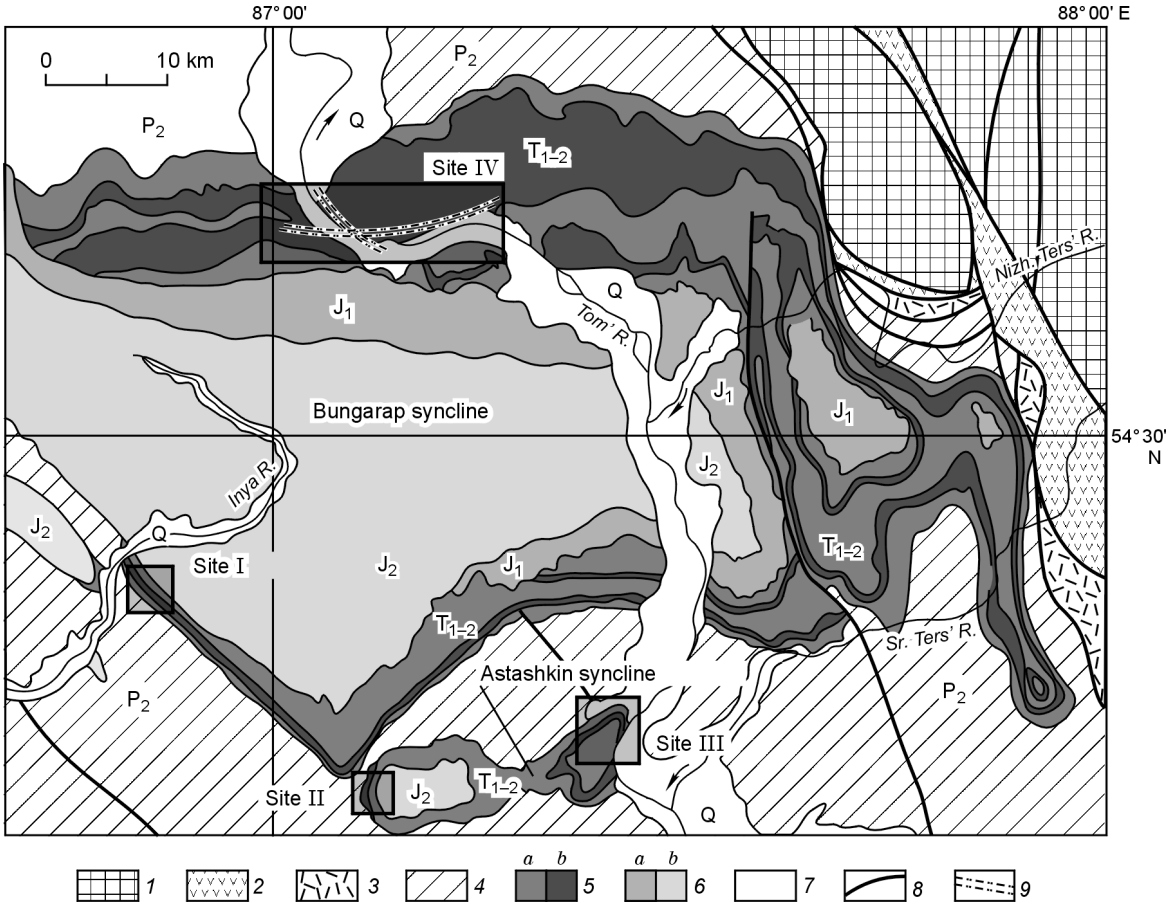



**Fig. 2. A scheme of geologic structure of the study area. *1–3* — Kuznetsk Alatau: *1* — undifferentiated complexes of Late Proterozoic-Early Paleozoic, *2* — Cambro-Ordovician volcanic complexes, *3* — Early Devonian volcanosedimentary complexes; *4–6* — Kuznetsk Basin: *4* — Permian molassa complexes, *5* — Triassic trap formation: tuffaceous sedimentary formations (*a*), bodies of basalts and microdolerites (*b*), *6* — complexes of filling of superposed depressions of: *a* — Early Jurassic, *b* — Middle Jurassic; *7* — Quaternary alluvium deposits; *8* — main rupture disturbances; *9* — zones of fracturing and secondary zeolitization.**

**Site II** is in the west of the Astashkin brachysyncline in two neighbor quarries in the vicinity of Kyrgai Village (54.25° N, 87.1° E). Basalts and siltstones from the zone of "hot" contact were sampled.

**Site III** is within the Astashkin brachysyncline. There, also two sections were studied, situated along the left bank of the Tom' River, near Ust'-Naryk Village. In one of them (54.3° N, 87.42° E), two subtabular basalt bodies were sampled. In the other section, 500 m downstream, samples were taken from the Upper Permian(?) underlying sedimentary rocks, represented there by massive sandstones and siltstones, in places obliquely laminated, dark-brown and greenish-gray. The contact between the sediment and basalts is not exposed.

**Site IV** lies in the north of the Bungarap syncline within the Saltymakovsky Ridge along the Tom' River. Nearly the whole volcanogenic part of the section of the Mal'tsev Group was sampled there across the strike. From the mouth of the Dulepova Brook to the Bugas Brook (54.67–54.70° N, 87.03–87.21° E), the sampled section is nearly completely made up of bodies of basalt and diabase porphyrites and microdolerites of black or dark-gray color, massive, rarely amygdaloidal, fine crystalline, composed of glass, laths of basic plagioclase, grains of clinopyroxene and an admixture of ore mineral, amounting in places to 10–12%. The rocks near the Bugas and Turala Brooks are represented by massive weakly altered varieties, whereas near the Moshkovo and Dulepov Brooks and the Bych'ya River, in the zone of fracturing and secondary zeolitization (see Fig. 2) the rocks are considerably altered.

## TECHNIQUE AND RESULTS OF PALEOMAGNETIC STUDIES

To identify the component composition of natural remanent magnetization (NRM), the samples were subjected to multistep temperature demagnetization and/or AF-demagnetization. Laboratory experiments were carried out using the equipment of the Paleomagnetic Center of the Institute of Geology, Novosibirsk, Russia, and Paleomagnetic Laboratory of the California University, Santa Cruz, USA. To measure the direction and magnitude of the vector of natural remanent magnetization, we used JR-4 and JR-5 spin-magnetometers, 2G Enterprises freezing magnetometer (USA) with a built-in unit of ac magnetic field. Depending on magnetization carrier, thermal demagnetization to temperatures of 500–700 °C was performed at a step of 30–40 °C, with the use of thermal demagnetizing device of UCSC design (USA) and V.P.Aparin's shielded furnace. To clear up the composition of minerals carrying magnetization, we used SIRM analysis. The results of laboratory experiments were analyzed by standard techniques [17] with the use of specialized software [18, 19].

The magnitudes of natural remanent magnetization (NRM) and magnetic perceptibility of basalts are generally rather high: more than 1000 mA/m and over $10^{-3}$ SI units, respectively. Factor $Q_n$ is about or much higher than 10, which is typical of unaltered rocks with thermal remanence [20].

The most typical Fe-bearing ore minerals in the basalts of the Siberian trap province are primary magmatic Ti-magnetites and products of their heterophase oxidation (solid exsolution), chiefly, magnetite [21]. Results of thermomagnetic analysis of remanent magnetization saturation SIRM($T$) of samples of Permian-Triassic basalts of the Kuznetsk Basin (Fig. 3) and corresponding deblocking temperatures ($T_{db}$) of stable magnetization components in the course of step-by-step thermal demagnetization are in agreement with mineralogical data on Siberian traps. The main fraction of remanent magnetization falls on the magnetic phase with $T_{db}$ of about 300–350 °C, but another magnetic phase with $T_{db}$ of 500–580 °C is also recorded. The basalts of the southern part of the studied area (sites I–III) are characterized by the coexistence of both phases in a sample. The NRM($T$) curves show a distinct bending in the interval of temperatures 300–350 °C, whereas demagnetization is completed at 500–580 °C (Fig. 4). The directions of the components destroyable within corresponding temperature ranges coincide within 1–3° and Zijderveld diagrams have a distinct one-component shape (see Fig. 4, *a*). We believe that relatively Ti-enriched Ti-magnetite (with $T_{db}$ of 300–350 °C) is a primary magnetic mineral, and the presence of the second, relatively Fe-enriched, phase (with $T_{db}$ of 500–580 °C) is due to heterophase oxidation of initial Ti-magnetites. It is known that the heterophase oxidation does not eliminate the paleomagnetic information recorded in grains of Ti-magnetite as their domain structure and magnetic state remain intact. On the contrary, the high stability of Ti-magnetites formed under heterophase oxidation favors paleomagnetic studies at such objects [22]. The composition of initial Ti-magnetites identifiable from deblocking temperatures of the basalts in the Kuznetsk Basin (on the assumption that these temperatures correspond to Curie's points) approximates 40% ulvospinel end-member in Ti-magnetite [23]. This, in turn, permits the depth of the magma source to be estimated at 35–50 km [24, 25], which agrees with the depth of magma sources in rift zones of continents [22].

The basalts of the northern region (site IV) are characterized, chiefly, by the presence in specimens of the only magnetic phase with $T_{db}$ of 300–350 °C or 500–580 °C. Zijderveld diagrams also have a one-component representation (see Figs. 4, *b*, *c*) with the exception of a viscous component demagnetized as early as first steps

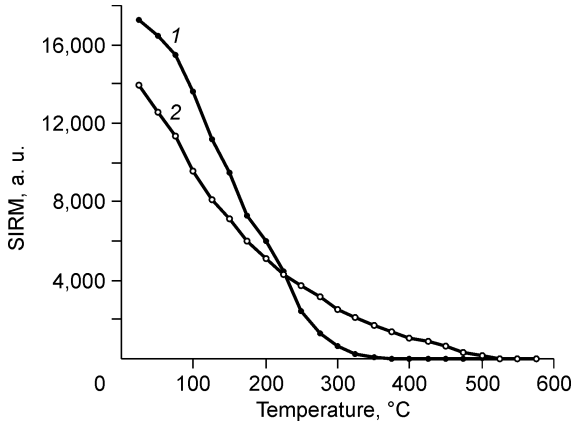



**Fig. 3. An example of SIRM analysis of a sample of the Permian-Triassic basalts of the Kuznetsk Basin. Curves of first (*1*) and second (*2*) heating.**

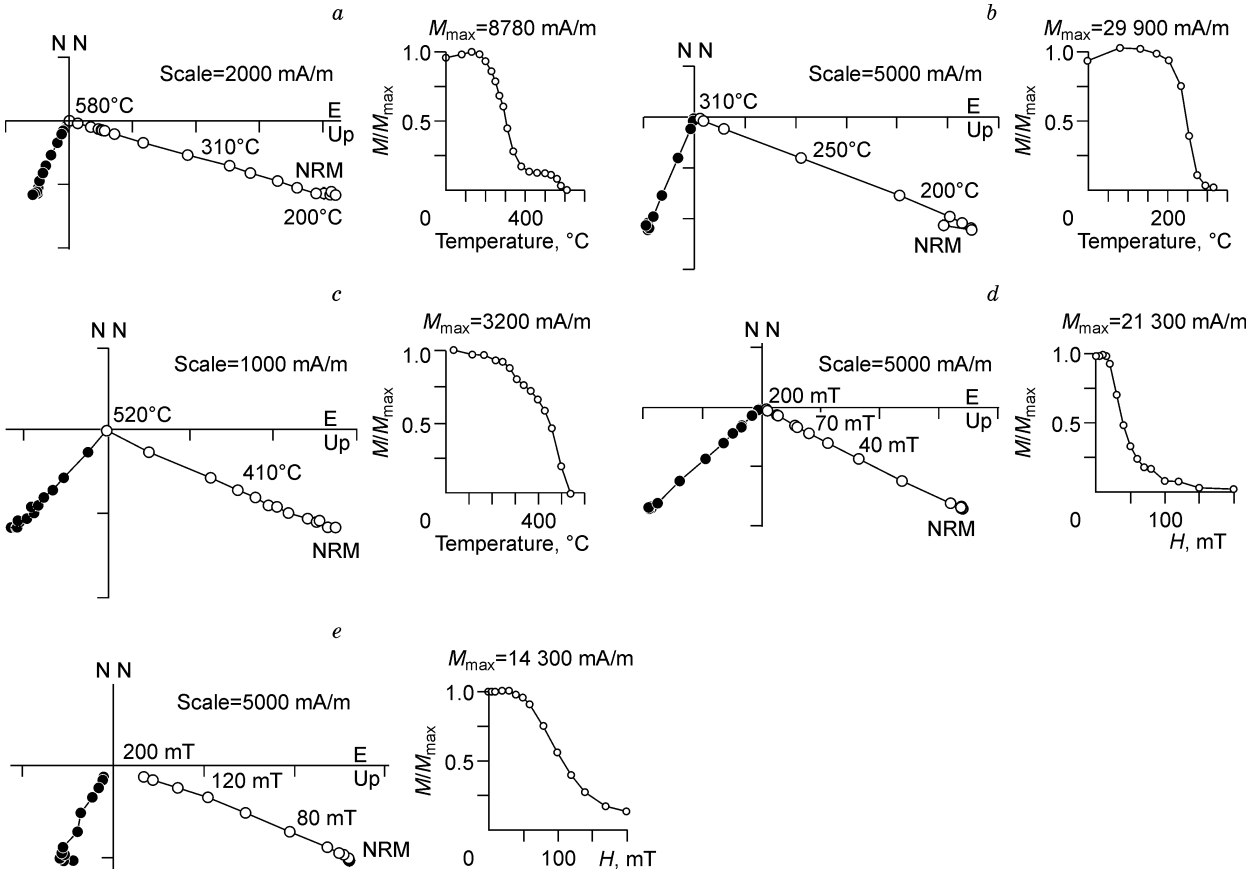



**Fig. 4. Characteristic Zijderveld diagrams and NRM(*T*, *H*) plots according to results of stepped thermal and AF-demagnetization of the Permian-Triassic basalts of the Kuznetsk Basin. *a–e* — explanations in the text. Hereafter, on the Zijderveld diagrams the filled circles designate vector projections on the horizontal plane, and open circles, those on the vertical plane. The diagrams are given in the stratigraphic coordinates.**

of heating. The directions of characteristic components related to a certain magnetic phase coincide within an error of detection.

Depending on the composition of Ti-magnetites present in the specimens or percentage ratio of initial or

oxidized Ti-magnetites, the magnitude of median demagnetizing field (MDF) changes on AF-demagnetization. When the sample is dominated by Ti-enriched Ti-magnetite, MDF is about 30 mT, otherwise MDF is essentially higher, up to 70 mT. Irrespective of this, Zijderveld diagrams on AF-demagnetization as well as on thermal demagnetization are one-component except small viscous component (see Fig. 4, *d*, *e*).

For scarce specimens whose sampling points structurally coincide with a zone of fracturing and secondary superimposed zeolitization in the north of the Bungarap syncline (see Fig. 2), small variations are observed in values of median demagnetizing fields and deblocking temperatures. These rocks are also characterized by maximum variations in petromagnetic parameters. The poorest preservation of natural remanent magnetization is also recorded there. These specimens have, at least, two components. One is of direct polarity close in direction to the modern geomagnetic field. The stability of this component is rather high. The spectra of blocking temperatures and coercitive spectra of this component are considerably overlapped with those of the other, the characteristic component of reversed magnetization. Therefore, to separate these components, we used in some cases the method of remagnetization circles.

## DATING OF MAGNETIZATION COMPONENTS

To date the magnetization components relative to geologic events, baked-contact and fold tests were used.

The underlying Upper Permian sediments sampled in the south of the area (sites I and III, see Fig. 2) are represented by variegated sandstones and siltstones. The rocks are characterized by low values of natural remanent magnetization (about 1 mA/m). The direction of magnetization is difficult to determine. Because of low NRM the vector diagrams of thermal demagnetization are characterized by a great scatter of directions (Fig. 5, *a*). At three sampling points we succeeded in obtaining characteristic directions only for seven samples (Table 1). Total demagnetization is reached at temperatures no more than 600 °C and is, possibly, related to the presence of a small amount of magnetite in the samples. Despite low statistical parameters of distribution, $K_s/K_g$ = 8.5/5.3 at $N$ = 7, the fold test [26] indicates the prefolding age of the isolated component. The maximum precision is reached on unfolding, 81.2%. In this case, the 95% confidence interval, from 66.4 to 96.6%, though does not cross, approaches "complete unfolding".

Unlike the unbaked sediments, the samples of the exocontact of one of the explored basalt tabular bodies in the south of the Bungarap syncline (point KG-1) show a distinct characteristic component of magnetization (ChRM). In the samples taken in the immediate vicinity of the contact (≤10 cm), the characteristic component destroys at over 630 °C, which suggests hematite as a potential carrier of ChRM (see Fig. 5, *b*). At a greater distance from the contact (to 0.5 m) the characteristic component is also well expressed, but completely destroys at a temperature as low as 400 °C (see Fig. 5, *c*). As shown above, hematite was not established in unbaked Permian sediments on thermal demagnetization. Moreover, this mineral is not generally typical of coaliferous deposits [27]. This suggests that, in the immediate vicinity of the contact, hematite is of secondary nature. Most likely, it was produced by magnetite oxidation on baking, which is typical of surface effusions [22]. Thus, the time of formation of hematite and the time when it acquired magnetization correspond to the time of basalt formation, and the acquired ChRM is total thermal remanent magnetization (TRM). Away from the contact the baking temperature drops; as a result, hematite does not form, and the magnetic minerals present in the rock acquire partial thermal remanent magnetization (PTRM).

The directions of TRM and PTRM of the Permian rocks in the baking zone do not statistically differ from the basalt ChRM direction (angle between middle directions of 1.5±4.4°) but considerably differ from the direction of unbaked rocks. Moreover, the baked and unbaked rocks are of different polarity (Fig. 6). Thus, the baked-contact test performed in a classical manner is positive, i.e., there was no total regional remagnetization of the examined rocks after traps had formed. Accordingly, we can suppose that the magnetization of basalts corresponds to the time of their formation.

The precision of scatter of tilt-corrected middle directions in basalts is much higher than *in situ* ($K_s/K_g$ = 95.6/22.2 at $N$ = 28), which suggests the prefolding age of this component (see Fig. 6). The simulation parametric fold test [26] shows that the maximum precision is reached with 117.7% unfolding at a 95% confidence interval from 118.5 to 107.5%, i.e., with complete unfolding. The correlation fold test [28] also gives a positive result (DC Slope: 1.133±0.197).

Thus, we can surely state that the ChRM component recorded in the basalts is due to the Earth's magnetic field at the moment of formation of the Kuznetsk traps. This permits us to use the paleomagnetic directions imprinted in the traps equally to reconstruct the paleogeographic position of the structure at the Permian-Triassic boundary and to reveal synchronous levels of intraplate magmatic activity on the territory of the Siberian Platform, Taimyr, and West Siberia.

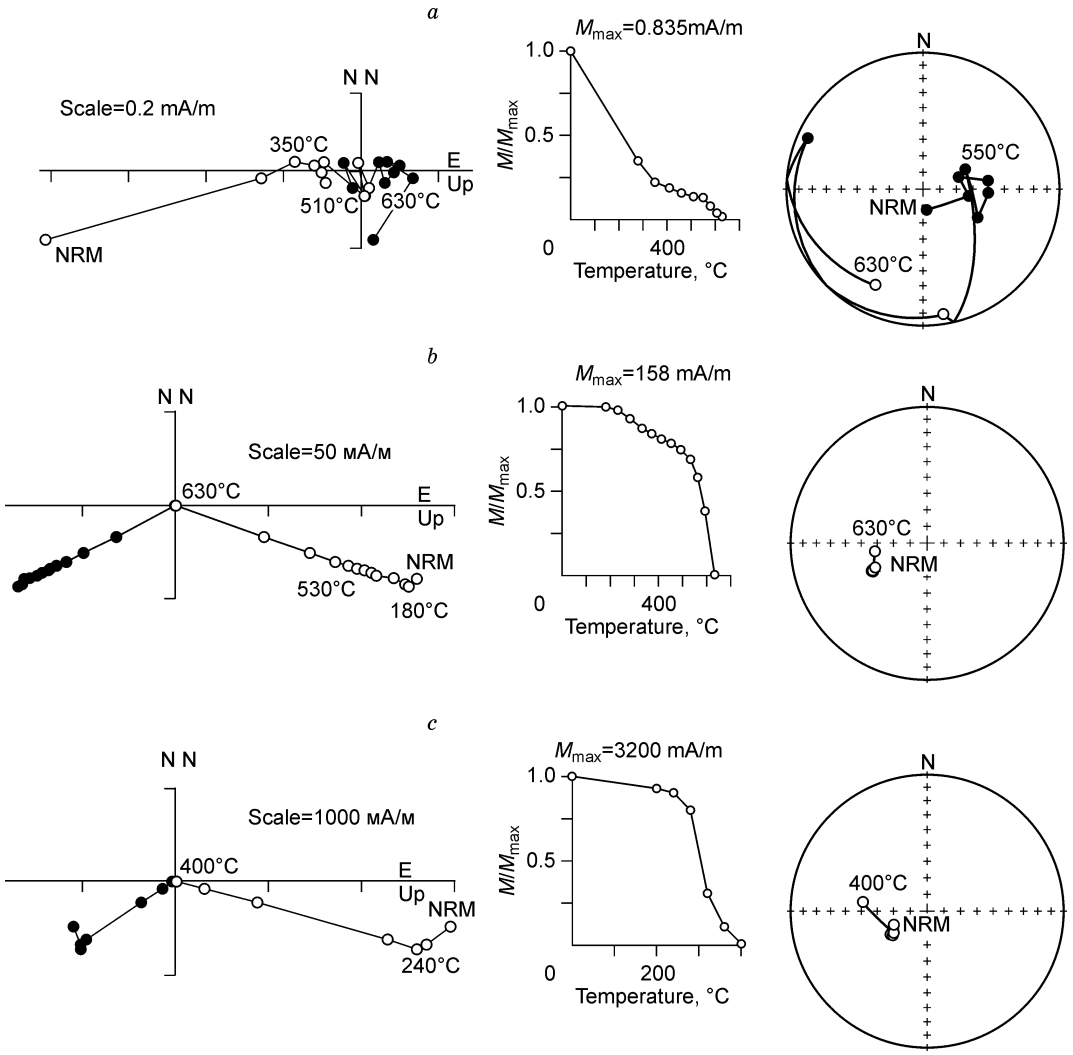



**Fig. 5. Characteristic Zijderveld diagrams, NRM(*T*) plots and stereograms on the basis of results of stepped thermal demagnetization of the Permian sedimentary rocks of the Kuznetsk Basin. *a* — Unbaked sandstones, *b* — contact sandstones, *c* — sandstones from the baked zone. Hereafter, the filled symbols on stereograms designate projections of vectors on the lower hemisphere, and the open symbols, those on the upper hemisphere.**

## DISCUSSION. TECTONIC AND GEODYNAMIC IMPLICATIONS

The most important and remarkable fact is that magnetization of all examined basalts, which generally characterize quite a complete section of the Kuznetsk traps 1500 m in total thickness, have only reversed polarity. As to the Permian underlying sediments, the directions of their ChRM are characterized, on the contrary, by direct polarity. At the same time it is well known that the whole trap formation of the Siberian Platform has a direct polarity and only the lowermost horizons of the Ivakino Formation are magnetized reversely [29]. According to data of isotope geochronology, the age of basalts of the Ivakino Formation is 248.5±2.4 Ma [8]. A similar situation is in Western Taimyr, where the reversed polarity has been established only in the deposits of the Syradasai Formation making up the lower part of the section [30]. The most complete paleomagnetic section of the Permian-Lower Triassic is based on the data obtained from superdeep BH-6 in the Urengoi-Koltogory graben of the rift system of the West Siberian Plate [31, 32]. According to paleomagnetic data, this volcanosedimentary section of the West Siberian traps was formed from Upper Tara time of the Late Permian to Olenek time of the Early Triassic (in the interval 250–245 Ma) [32]. The age analog of the Mal'tsev Group of the Kuznetsk Basin

**Table 1**

**Paleomagnetic Directions in Permian-Triassic Basalts and Permian Sedimentary Rocks of the Kuznetsk Basin**

| Object | *N* | $D_g$ | $I_g$ | $D_s$ | $I_s$ | *K* | $\alpha_{95}$ |
|---|---|---|---|---|---|---|---|
| | | | *Site* I (*Karakan Ridge*) | | | | |
| 98s001 | 13*z* | 246.6 | –85.1 | 219.2 | –63.7 | 86.9 | 4.5 |
| 98s002 | 2*z* + 6*c* | 216.4 | –77.6 | 214.3 | –55.6 | 199.2 | 4.3 |
| 98s003 | 9*z* | 79.7 | –79.1 | 202.2 | –75.9 | 54.0 | 7.1 |
| 98s004&5 | 11*z* | 162.5 | –79.7 | 206.9 | –62.1 | 129.3 | 4.0 |
| ad-6 | 14*z* | 217.2 | –86.2 | 225.5 | –64.2 | 360.7 | 2.1 |
| ad-7 | 10*z* | 41.4 | –87.6 | 227.7 | –70.4 | 379.1 | 2.5 |
| | | 176.6 | –86.5 | — | — | 94.4 | 6.9 |
| Object mean for site I | | — | — | 216.4 | –65.5 | 102.9 | 6.6 |
| | | VGPole: Lat = 66.8, Long = 176.7; *dp*/*dm* = 10.7/8.7; Plat: 47.7 ± 9.7 | | | | | |
| | | | *Site* II (*near Kyrgai Village*) | | | | |
| 98s018 | 3*z* | 118.7 | –77.9 | 222.5 | –60.3 | 116.1 | 11.5 |
| 98s019 | 5*z* | 156.8 | –78.3 | 228.5 | –63.2 | 866.4 | 2.6 |
| KG-1 | 9*z* + 2*c* | 113.4 | –67.4 | 223.7 | –62.6 | 70.2 | 5.5 |
| | | 125.5 | –75.3 | — | — | 91.2 | 13.0 |
| Object mean for site II | | — | — | 224.8 | –62.1 | 1448.1 | 3.2 |
| | | VGPole: Lat = 59.1, Long = 179.1; *dp*/*dm* = 5.0/3.9; Plat: 43.4 ± 4.4 | | | | | |
| | | | *Site* III (*near Ust'-Naryk Village*) | | | | |
| 98s010&11 | 5*z* | 217.9 | –64.6 | 217.9 | –64.6 | 252.2 | 4.8 |
| 98s012&13 | 4*z* | 218.8 | –60.3 | 218.8 | –60.3 | 649.3 | 3.6 |
| 98s014 | 4*z* | 222.2 | –64.7 | 222.2 | –64.7 | 257.2 | 5.7 |
| 98s015 | 4*z* | 228.7 | –63.1 | 228.5 | –63.2 | 236.8 | 6.0 |
| 98s016 | 6*z* | 234.7 | –65.9 | 234.7 | –65.9 | 227.5 | 4.5 |
| | | 224.2 | –63.9 | — | — | 460.1 | 3.6 |
| Object mean for site III | | — | — | 224.2 | –63.9 | 460.1 | 3.6 |
| | | VGPole: Lat = 60.7, Long = 176.2; *dp*/*dm* = 5.7/4.6; Plat: 45.6 ± 5.1 | | | | | |
| | | | *Site* IV (*Saltymakovsky Ridge*) | | | | |
| ad-3 | 4*z* + 7*c* | 241.8 | –75.3 | 248.0 | –63.5 | 173.4 | 3.6 |
| 97s007 | 2*z* + 3*c* | 215.5 | –74.8 | 232.7 | –64.5 | 180.7 | 6.5 |
| 97s008 | 2*z* + 2*c* | 211.3 | –77.7 | 233.0 | –67.6 | 87.0 | 11.5 |
| 97s005&6 | 2*z* + 3*c* | 216.8 | –76.6 | 234.9 | –66.1 | 75.2 | 10.1 |
| 97s009 | 2*z* + 1*c* | 185.8 | –80.1 | 224.3 | –72.1 | 69.3 | 17.5 |
| 97s010 | 1*z* + 2*c* | 156.4 | –76.1 | 200.3 | –73.3 | 706.1 | 7.7 |
| 97s011&12 | 9*z* | 188.6 | –71.2 | 212.9 | –64.2 | 94.6 | 5.3 |
| ad-5 | 12*z* | 211.3 | –65.8 | 233.5 | –65.3 | 163.0 | 3.4 |
| 97s001 | 10*z* | 248.8 | –48.7 | 244.0 | –55.8 | 275.6 | 2.9 |
| 97s002 | 3*z* | 253.4 | –49 | 249.6 | –56.4 | 305.7 | 7.1 |
| 97s003 | 5*z* | 254.2 | –47.6 | 250.5 | –55.1 | 848.8 | 2.6 |
| 97s004 | 5*z* | 241.8 | –41.8 | 237.3 | –48.4 | 73.5 | 9.0 |
| ad-1 | 13*z* | 207.2 | –65.3 | 212.7 | –67.0 | 354.5 | 2.2 |
| ad-2 | 7*z* | 207.2 | –66.7 | 213.2 | –68.4 | 620.9 | 2.4 |
| | | 227.3 | –67.8 | — | — | 22.9 | 8.5 |
| Object mean for site IV | | — | — | 233.0 | –64.2 | 70.3 | 4.8 |
| | | VGPole: Lat = 56.8, Long = 169.1; *dp*/*dm* = 7.7/6.1; Plat: 46.0 ± 6.9 | | | | | |
| Mean of all objects | | 218.4 | –73.9 | — | — | 22.2 | 5.9 |
| | | — | — | 227.1 | –64.3 | 95.8 | 2.8 |
| | | VGPole: Lat = 60.0, Long = 172.7; *dp*/*dm* = 4.5/3.6; Plat: 46.1 ± 4.0 | | | | | |
| | | | *Permian sedimentary rocks* | | | | |
| 98s006. | 008-009 | 7*z* | 34.9 | 67.4 | — | — | 5.3 |
| | | | — | 21.9 | 66.7 | 8.5 | 21.9 |

**Note. *N* — number of samples (*z* — component analysis, *c* — circles of remagnetization); $D_g$ — declination in geographical *(in situ)* and $D_s$, in stratigraphical (tilt-corrected) coordinates; $I_g$ — inclination in geographical and $I_s$, in stratigraphical coordinates; *K* — precision; $\alpha_{95}$ — radius of confidence oval; VGPole — coordinates of paleomagnetic North Pole; Lat — latitude (in degrees of N latitude); *dp*/*dm* — semiaxes of confidence oval for mean pole; Plat — paleolatitude at sampling locality.**

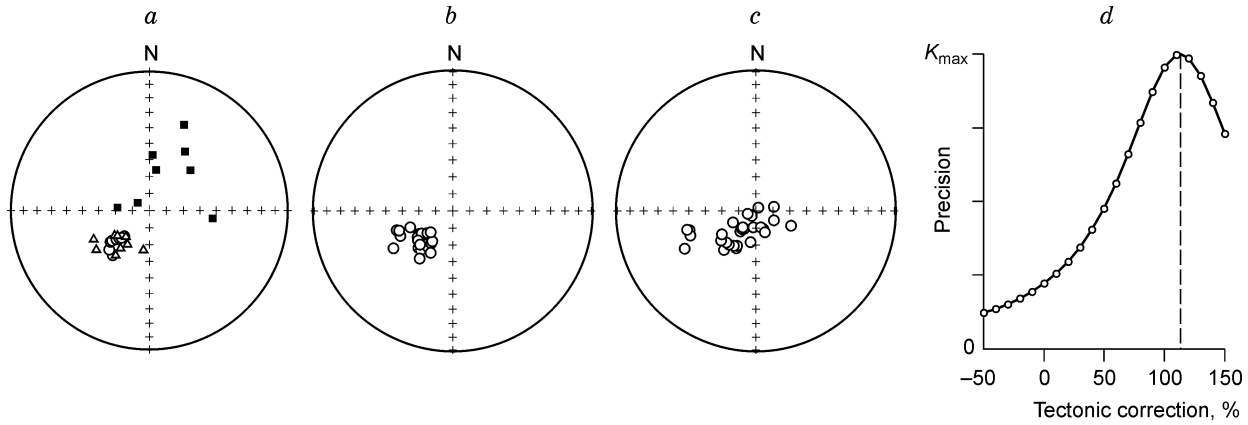



**Fig. 6. Results of paleomagnetic tests. *a* — Baked-contact test: circles show distribution of the tested directions of basalt magnetization near Kyrgai Village (site II, point 98s018 and 98s019), triangles show directions of magnetization of baked host rocks (point KG-1), squares stand for directions of magnetization for unaltered Permian sedimentary rocks far from the zone of baking near the Karakan Ridge (site I, point 98s006) and near Ust'-Naryk Village (site III, points 98s008 and 98s009); *b–d* — fold-test results: *b* — tilt-corrected distribution of site-mean directions of ChRM of basalts, *c* — the same, *in situ*, *d* — plot of change of precision of distribution under unfolding.**

seems to be inversely magnetized rocks of the middle part of the Aimal Formation (see Fig. 7). According to Kazansky et al. [32], the Permian-Triassic boundary runs through the roof of the zone of reversed polarity in the middle part of the Aimal Formation. Comparison of magnetostratigraphic sections of the trap formation of West Siberia, Noril'sk district, Taimyr and Kuznetsk Basin (see Fig. 7) with the "Magnetostratigraphic scale of the USSR" [33] and "Composite magnetostratigraphic scale" [34] permits more precise dating of the formation of traps of the Kuznetsk Basin and general estimation of the duration and intensity of trap magmatism in the Siberian region.

The supposed correlation implies that the trap formation began quite synchronously throughout Siberia, from Kuznetsk Basin to Western Taimyr [2]. Moreover, new paleomagnetic and isotope-geochronological data on the Semeitau volcanoplutonic structure of the Chara zone (reversed magnetization of basalts and vitrophyres with $^{40}Ar/^{39}Ar$ age of 248.8±0.5 Ma) [35] suggest that volcanic activity proceeded synchronously in Eastern Kazakhstan as well. The synchroneity of tectonomagmatic activity on such a large territory may be due to nothing else but the influence of a single mantle superplume [1]. However, the duration and its intensity of this event in separate terranes are rather different and are owing, probably, to a certain suite of regional and geodynamical factors. The time of formation of inversely magnetized rocks of the Mal'tsev Group corresponds to the middle part of the Aimal Formation of West Siberia, Ivakino Formation of the Noril'sk district and Syradasai Formation of Western Taimyr, as well as to the upper part of the $R_3P$ magnetozone of the Illavara hyperzone (see Fig. 7). Our data suggest that the formation of traps in the Kuznetsk Basin at the Permian-Triassic boundary as compared with the rest of Siberia was the shortest event. The whole studied section in time of formation covers no more than 1 mln years at the Permian-Triassic boundary. The paleomagnetic ages of the Kuznetsk basalts are in agreement with $^{40}Ar/^{39}Ar$ isotope data — 249±2 [36] and results of petrologo-geochemical studies, which reveals a similarity between the Kuznetsk traps and initial basalts of trap formations of the Siberian Platform and basement of the West Siberian Plate [37]. The volcanic activity on the territory of West Siberia was much longer but less intense than in other regions. On the basis of correlation of the paleomagnetic section of superdeep BH-6, the time of formation of the traps even in this most complete section did not exceed 5–6 Myr [32]. In the sections of trap formation in the northern Siberian Platform, which are much longer than in the other areas, the number of magnetozones in a section does not exceed three. Accordingly, the duration of trap magmatism on the Siberian Platform can be reduced to 2 Myr and less. The comparison suggests that it is during this period in the earliest Triassic that the whole directly magnetized trap sequence of the Noril'sk district and the rocks of the Labak Formation of Western Taimyr could be formed (see Fig. 7). At that time, magmatism seemed to terminate within the Kuznetsk Basin.

Thus, widespread geologic bodies formed on the vast territory for a short period of time are the most profitable

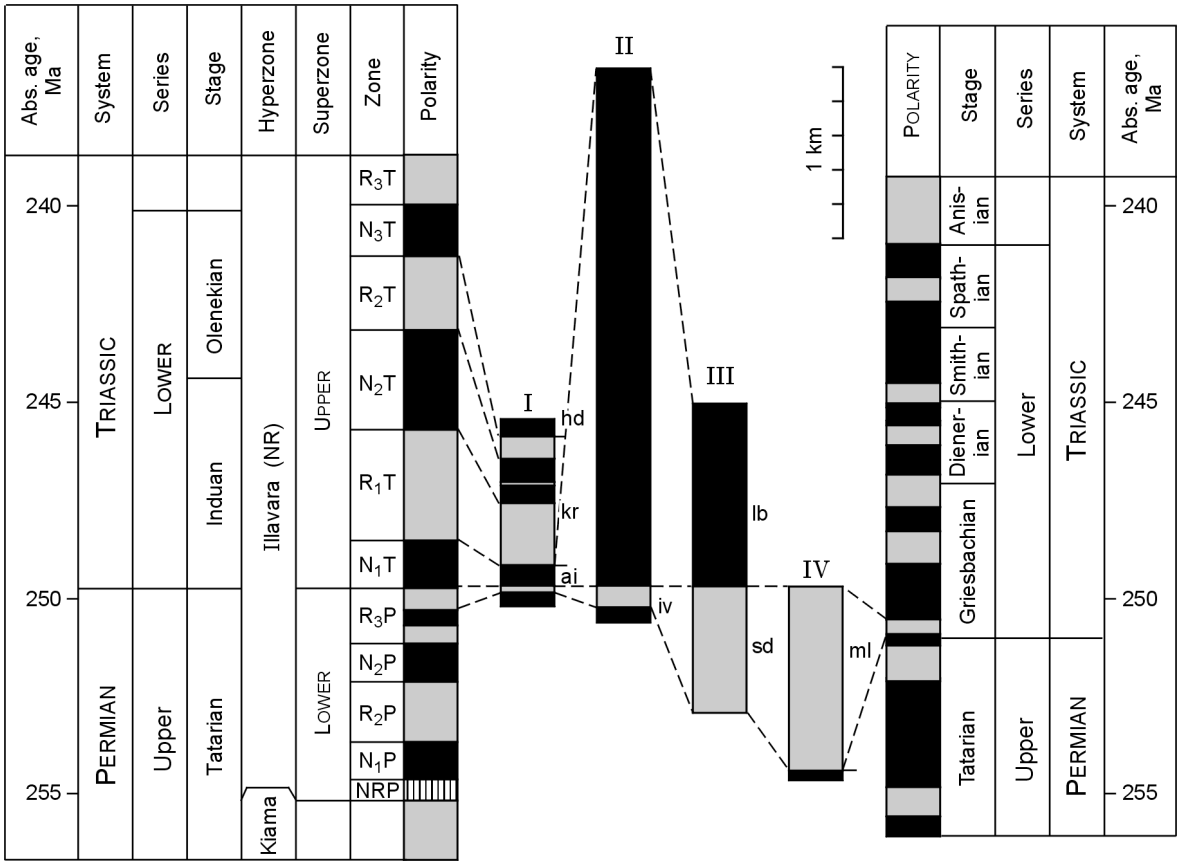



**Fig. 7. Comparison of magnetostratigraphic sections of trap complexes of Siberia with the Magnetostratigraphic Scale of the USSR (left), after [33], and Composite Magnetostratigraphic Scale (right), after [34]. I — superdeep BH-6, West Siberian Plate [32], II — Noril'sk district, Siberian Platform [29], III — Western Taimyr [30], IV — Kuznetsk Basin [this paper]. Letters at corresponding intervals of magnetostratigraphic sections designate local stratigraphic subdivisions — formations and groups: ai — Aimal, kr — Korotchaevo, hd — Khadyr'yakhs, iv — Ivakino, sd — Syradasai, lb — Labak, ml — Mal'tsev. The interval of direct polarity (black) in the magnetostratigaphic section of the Kuznetsk Basin corresponds to the magnetization of trap-underlying sedimentary rocks of Late Permian age.**

object for an exact estimate of the extent and kinematics of within-plate movements of the Siberian Platform and its framing in the early Mesozoic.

Analysis of paleomagnetic database [38] revealed a significant discrepancy between paleomagnetic directions of traps of the Kuznetsk Basin and the Siberian Platform. Figure 8 shows a distribution of the available magnetization directions in the late Permian-early Triassic (260–240 Ma). They were determined chiefly from the rocks of trap formation of the Siberian region. Obviously, the orientation of the paleolatitudes of the Siberian Platform, Taimyr, and central parts of the Altai-Sayan region (including the Minusa system of depressions) [39] is the same and, in the present framework of references, is directed submeridionally. Most likely, of the same orientation should be the Permian-Triassic paleolatitudes within the West Siberian Plate, at least, in its eastern part. The paleolatitude (47° N) determined for the basalts from superdeep BH-6 [32] is well correlated with the paleolatitudes of the Siberian Platform. At the same time, the paleolatitudes of traps of the Kuznetsk trough are oriented at an angle to the paleolatitudes of the Siberian Platform (see Fig. 8). The angular distance between the position of middle paleomagnetic poles of the Kuznetsk Basin (PLat = 60.0, PLong = 172.7, $\alpha_{95}$ = 4.0) and Siberian Platform (PLat = 49.0, PLong = 151.6, $\alpha_{95}$ = 5.0) is 24±7.2° of the arc of a great circle. The most likely explanation is a strike-slip movement of the Kuznetsk structures relative to the southwestern margin of the Siberian Platform during or after trap formation. This rotation mechanism was considered by Khramov [40] in addition to the mechanisms described in [41]. The strike-slip movement occurred, most probably, along a series of arcuate

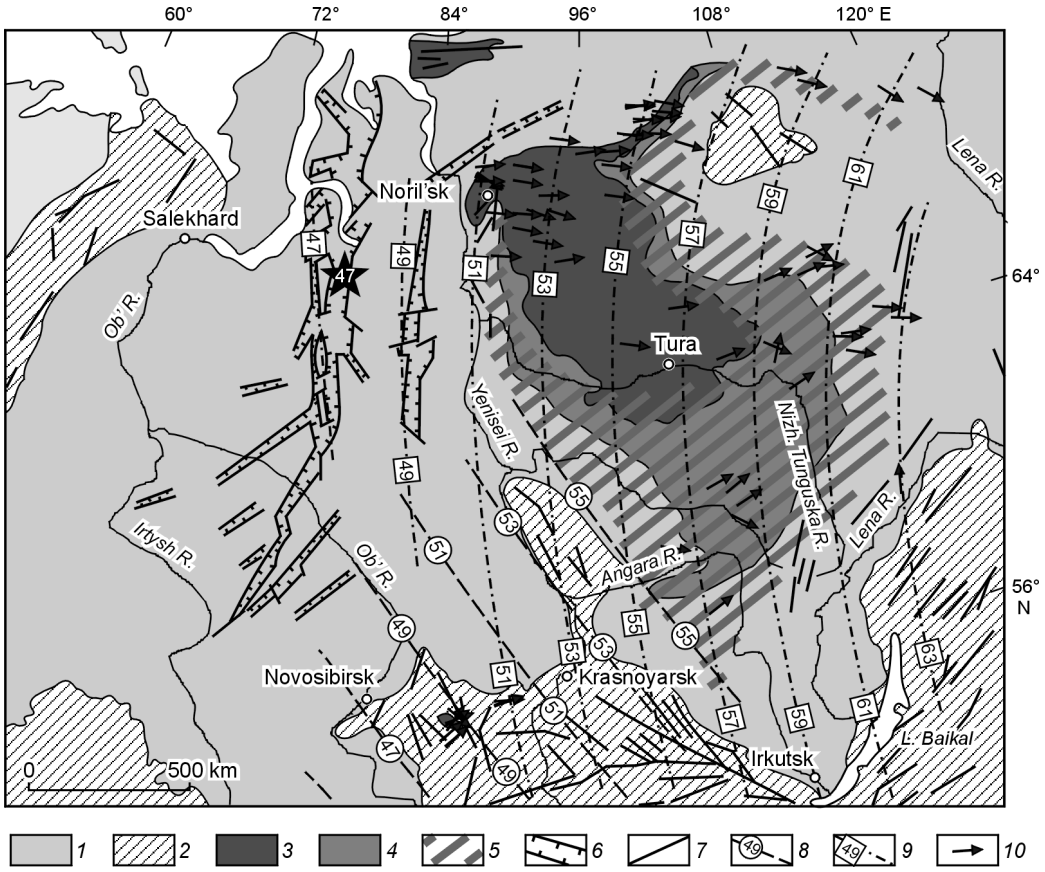



**Fig. 8. Schematic distribution of Permo-Triassic traps of Siberia (after [2], simplified and supplemented) and position of Permo-Triassic paleolatitudes. *1* — Phanerozoic sedimentary cover; *2* — foldbelts and inliers of the Precambrian basement; *3–5* — trap formation: *3* — basalts, *4* — basalt tuffs and tuffites, *5* — areas of development of intrusive traps; *6* — rift structures of the pre-Jurassic basement of the West Siberian Plate; *7* — main tectonic disturbances; *8* — paleolatitudes determined from results of analysis of the Permo-Triassic magnetization directions of the Kuznetsk traps; *9* — the same, for East Siberia, on the basis of paleomagnetic data [38]; *10* — Permo-Triassic paleomagnetic directions. Starred is the paleolatitude determined from the core of superdeep BH-6 [32].**

zones of strike-slip faults, including the Kuznetsk-Altai system of faults separating the Kuznetsk Basin from the central part of the Altai-Sayan region. Slippage and lagging of the Kuznetsk structures relative to the Siberian Platform when it moves with a clockwise rotation led to deformations and, consequently, to a turn of the paleolatitudes [42]. In this case, the Permian-Triassic volcanotectonic synclines of the Kuznetsk Basin formed according to the pull-apart scheme [43].

The Mesozoic strike-slip faults that occurred under the conditions of general sublatitudinal (in the recent framework of reference) compression in the southwestern Siberian Platform were chiefly caused by a difference in angular velocities of movement of Siberia and Eastern Europe. As inferred from global reconstructions [44, 45] and available paleomagnetic data, the main tectonic blocks of Northern Eurasia were a single continental mass as early as the Early Mesozoic. The continent had a submeridional orientation, and its Siberian part was in the high latitudes of the Northern Hemisphere (not farther southward than 40° N). The movement of the Eurasian Plate in this interval is a gradual clockwise turn. However, East Siberia moved faster than Eastern Europe. As a result of advancing, conditions of stretching appeared within northern West Siberia, whereas the constriction setting was dominated in the south of Siberia [13, 40]. A consequence was the formation of graben structures of the West Siberian rift system, on the one hand, and deformations and considerable intraplate strike-slip movements (including those within the southwestern framing of the Siberian Platform), on the other. Kinematically close strike-slip

movements within the Asian continent might, probably, continue until the end of the Mesozoic. Evidence comes from new paleomagnetic data on the Late Jurassic—Early Cretaceous of Transbaikalia, which testify to the presence of strike-slip deformations of sinistral kinematics in the south of the Siberian Platform [46].

## CONCLUSIONS

1. Magnetization of the rocks of trap formation of the Kuznetsk Basin is characterized only by reversed polarity, the time of its formation corresponds to the upper part of the $R_3P$ magnetozone of the Illavara hyperzone, i.e., to the Permian-Triassic boundary (ca. 250 Ma). The duration of magmatism in this region seemed to be no more than 1 Myr.

2. According to paleomagnetic data, the trap formation began quite synchronously in all regions of Siberia from Kuznetsk Basin to Western Taimyr, which may be due to the influence of a single superplume. However, the duration of this event and its intensity in each separate region are different and are controlled by regional and geodynamic agencies.

3. Considerable differences have been established in the orientation of paleomeridians and paleolatitudes of the Kuznetsk Basin and Siberian Platform, which reflect processes of intraplate tectonic activity. The most probable explanation is the sinistral strike-slip movement and deformation of the structures of the Kuznetsk Basin caused by "lagging" of this tectonic unit during the general movement and clockwise rotation of the Siberian craton "inside" the Eurasian Plate. The magmatic and volcanic activity within the Kuznetsk Basin at the Permian-Triassic boundary is related to the formation of "pull-apart"-type depressions near strike-slip faults.

We thank Robert Coe and Xixi Zhao of Californian University in Santa Cruz for the possibility of using the equipment of the Paleomagnetic Laboratory of the University and assistance in carrying out the experiments.

This work was supported by integrated projects of the Siberian Branch of the RAS 6.7.4 and 134, by grant 04-05-64363 from the Russian Foundation for Basic Research, by the Presidium of the SB RAS, Foundation for National Science Support, grant MK-4334.2004.5 from the Foundation of the President of the Russian Federation, as well as by grant EAR 98-054 from the National Science Foundation of the USA.

## REFERENCES

1. Dobretsov, N.L., Permian-Triassic magmatism and sedimentation in Eurasia as a reflection of superplume, *Dokl. RAN*, **354**, 2, 220–223, 1997.

2. Al'mukhamedov, A.I., A.Ya. Medvedev, and N.P. Kirda, Comparative analysis of geodynamic settings of the Permo-Triassic magmatism in East Siberia and West Siberia, *Geologiya i Geofizika (Russian Geology and Geophysics)*, **40**, 11, 1575–1587(1550–1561), 1999.

3. Kutolin, V.A., *Trap formation of the Kuznetsk Basin* [in Russian], 117 pp., Izd. AN SSSR, Novosibirsk, 1963.

4. Aplonov, S.V., Paleogeodynamics of the West Siberian Plate, *Sov. Geologiya*, 7, 27–36, 1989.

5. Purtova, S.I., Detailed justification of the age of Early Mesozoic and Permian in the section of the Tyumen' superdeep borehole, in *The Tyumen' superdeep borehole (range 0–7502). Results of drilling and research. Issue 4: Research driling in Russia* [in Russian], 94–100, KamNIIKIGS, Perm', 1996.

6. Zolotukhin, V.V., and A.I. Al'mukhamedov, Trapps of Siberian Platform, in J.D. McDougal (ed.), *Continental Flood Basalts*, 273–310, Kluwer, 1988.

7. Baksi, A.K., and E. Farrar, $^{40}Ar/^{39}Ar$ dating of whole rock basalts (Siberian Trapps) in the Tungusska and Noril'sk areas, USSR, *EOS*, **72**, p. 570, 1991.

8. Dalrymple, G.B., G.K. Czamanske, G.K., V. Stepanov, V.A. Fedorenko, and M.A. Lanphere, $^{40}Ar/^{39}Ar$ ages from samples from the Noril'sk-Talnakh ore-bearing intrusions and the Siberian flood basalts, *ibid*., p. 570.

9. Renne, P.R., and A.R. Basu, Rapid eruption of the Siberian trapps flood basalts at Permo-Triassic boundary, *Science*, **253**, 176–179, 1991.

10. Campbell, I.H., G.K. Czamanske, V.A. Fedorenko, R.I. Hill, and V. Stepanov, Synchronism of the Siberian Trapps and the Permian-Triassic boundary, *Science*, **258**, 1760–1763, 1992.

11. Zolotukhin, V.V., V.A Ponomarchuk, A.V. Travin, Yu.V. Vasil'ev, and A.I. Al'mukhamedov, Evolution of Permian-Triassic magmatism in the northern Siberian Platform according to Ar/Ar data, in *Problems of petrology and minerageny of mafic-ultramafic complexes* [in Russian], issue 1 58–61, TGU, Tomsk, 1998.

12. Claoue-Long, J.C., Z. Zichao, V. Gougan, and D. Shaohua, The age of the Permian-Triassic boundary, *Earth Planet. Sci. Lett.*, **105**, 182–190, 1991.

13. Bazhenov, M.L., and A.A. Massakovsky, Horizontal movements of the Siberian Platform in the Triassic according to paleomagnetic and geological data, *Geotektonika*, 1, 59–69, 1986.

14. Zyatev, G.G., Z.A Myshko, and B.M. Perederin, Magnetic properties of Upper Permian-Lower Triassic rocks in the Kuznetsk Basin, in *Geology of the Kuznetsk Basin and Gorny Altai* [in Russian], issue 4, 179–180, Novokuznetsk, 1971.

15. Shokal'sky, S.P., G.A. Babin, A.G. Vladimirov, S.M Borisov, *Correlation of magmatic and metamorphic complexes of the western part of the Altai-Sayan folded area* [in Russian], ed. A.F. Morozov, 187 pp., Izd. SO RAN, filial "Geo", 2000.

16. *Geological map of the USSR. Scale of 1:1,000,000 (new series). Explanatory note. Novosibirsk. Sheet N-(44), 45* [in Russian], 134 pp., VSEGEI, Leningrad, 1988.

17. Butler, R.F., *Paleomagnetism: magnetic domains to geologic terrains* [in Russian], 319 pp., Backwell Sci. Publ. Oxford, 1992.

18. Enkin, R.J., *A computer program package for analysis and presentation of paleomagnetic data*, 16 pp., Pacific Geoscience Centre, Geological Survey of Canada, 1994.

19. Torsvik, T., and M. Smethurst, *GMAP v.32: Geographic mapping and palaeoreconstruction package*, 65 pp., Norges geologiske underskelse, NGU-rappor, 1998.

20. Mikhailova, N.M., A.M. Glevasskaya, and V.N. Tsykora, *Paleomagnetism of volcanogenic rocks and reconstruction of geomagnetic field of the Neogene* [in Russian], 200 pp., Naukova Dumka, Kiev, 1974.

21. Zolotukhin, V.V., A.I. Al'mukhamedov, and N.A. Tkachenko, Characteristics of the compositions of main rock-forming minerals of Deccan and Siberian traps (comparative aspect), in *Traps of Siberia and Deccan: similarities and differences* [in Russian], 140–177, IGiG SO AN SSSR, Novosibirsk, 1991.

22. Pechersky, D.M., and A.N. Didenko, *Paleoasian ocean* [in Russian], 298 pp., OIFZ RAN, Moscow, 1995.

23. Nagata, T., *Rock magnetism* [in Russian], 348 pp., Mir, Moscow, 1965.

24. Pechersky, D.M., V.I. Bagin, S.Yu. Brodskaya, and Z.V. Sharonova, *Magnetism and conditions of formation of igneous rocks* [in Russian], 288 pp., Nauka, Moscow, 1975.

25. Ermakov, V.A., and D.M. Pechersky, Nature of inclusions of gabbroids from young lavas of the Kurils, *Tikhookeanskay Geologiya*, 4, 45–55, 1989.

26. Watson, G.S., and R.J. Enkin, The fold test in paleomagnetism as a parameter estimation problem, *Geophys. Res. Lett.*, **20**, 2135–2137, 1993.

27. Erofeev, V.S., and Yu.G. Tsekhovsky, *Paragenetic associations of continental deposits: a family of humide parageneses* [in Russian], 210 pp., Nauka, Moscow, 1982.

28. Enkin, R.J., *Formation et deformation de l'Asie depuis la fin de l'ere primaire: les apports de l'etude paleomagnetique des formations secondaries de Chine du sud. PhD thesis*, 45 pp., University of Paris, 7, Paris, 1990.

29. Lind, E.N., S.V. Kropotov, G.K. Czamanske, S.C. Gromme, and V.A. Fedorenko, Paleomagnetism of Siberian flood basalts of the Noril'sk area: a constraint of eruption duration, *Intern. Geol. Rev.*, **36**, 12, 1139–1151, 1994.

30. Gurevitch, E.L., M. Westphal, J. Daragan-Suchov, H. Feinberg, and J.P. Pozzi, Paleomagnetism and magnetostratigraphy of the traps from Western Taimyr (Northern Siberia) and the Permo-Triassic crisis, *Earth Planet. Sci. Lett.*, **136**, 461–473, 1995.

31. Westphal, M., E.L. Gurevich, B.V. Samsonov, H. Feinberg, and J.P. Pozzi, Magnetostratigraphy of the Lower Triassic volcanics from deep drill SG in Western Siberia: evidence of long-lasting Permo-Triassic activity, *Geophys. J. Int.*, **134**, 254–266, 1998.

32. Kazanskii, A.Yu., Yu.P. Kazanskii, S.V. Saraev, and V.I. Moskvin, The Permo-Triassic boundary in volcanosedimentary section of the West Siberian plate according to paleomagnetic data (from studies of the Tyumenskaya superdeep borehole SDB-6), *Geologiya i Geofizika (Russian Geology and Geophysics)*, **41**, 3, 327–339(323–335), 2000.

33. Molostovsky, A.E., and A.N. Khramov, Paleomagnetic Phanerozoic scale and problems of magnetostratigraphy, in *Proceedings of the 27th IGC. Stratigraphy, section C01. Reports* [in Russian], vol. 1, 16–23, Nauka, Moscow, 1984.

34. Opdyke, N.D., and J.E.T. Channel, *Magnetic stratigraphy*, 346 pp., Academic Press, New York, 1996.

35. Lyons, J.J., R.S. Coe, X. Zhao, A.Y. Kazansky, A.E. Izokh, L.V. Kungurtsev, and D.V. Mitrokhin, Paleomagnetism of the Early Triassic Semeitau Igneous Series, E. Kazakhstan, *J. Geophys. Res.*, **107**, B7, 1010–1029, 2002.

36. Fedoseev, G.S., V.I. Sotnikov, A.F. Rubtsov, and V.O. Gimon, *Emendation of the Ob' subseries of the*

*West Siberian series of the legends of National Geological Maps-200: 2. Correlation of igneous complexes of the northern Kolyvan'-Tomsk folded zone* [in Russian], 342 pp., OIGGM SO RAN, Novosibirsk, 2000.

37. Kruk, N.N., A.V. Plotnikov, A.G. Vladimirov, and V.A. Kutolin, Geochemistry and geodynamic conditions of the formation of traps of the Kuznetsk Basin, *Dokl. RAN*, **369**, 6, 812–815, 1999.

38. McElhinny, M.W., and J. Lock, IAGA paleomagnetic database with Assecc. Surv., *Geophys*, **16**, 575–591, 1996 (http://dragon.ngu.no/Palmag/paleomag.htm).

39. Bragin, V.Yu., and D.V. Metelkin, Mesozoic strike-slip tectonics in the southwestern framing of the Siberian Platform as inferred from paleomagnetic study of basites, in *Proceedings of the Conference of Young Scientists in tribute to M.A. Lavrent'ev, 2001* [in Russian], Part II, 35–40, Izd. SO RAN, Novosibirsk, 2002.

40. Voronov, P.S. (ed.), *Role of strike-slip tectonics in the structure of the lithospheres of the Earth and terrestrial planets* [in Russian], 591 pp., Nauka, St. Petersburg, 1997.

41. Wells, R.E., and P.L. Heller, The relative contribution of accretion, shear and extension to Cenozoic tectonic rotation in the Pacific North West, *Geol. Soc. Amer. Bull.*, **100**, 325–338, 1988.

42. Kazansky, A.Yu. *Evolution of the structures of the western framing of the Siberian Platform according to paleomagnetic data. Abstract of ScD thesis* [in Russian], 40 pp., OIGGM SO RAN, Novosibirsk, 2002.

43. Berzin, N.A., A kinematic model for the formation of Altai-Sayan fold region: structural constraints, in *Continental Growth in the Phanerozoic: Evidence from Central Asia, Abstracts of Third Workshop IGCP-420*, 8–11, Izd. SO RAN, filial "GEO", Novosibirsk, 2001.

44. Zonenshain, L.P., M.I. Kuz'min, and L.M. Natapov, *Tectonics of lithosphere plates of the USSR territory* (in two books) [in Russian], Book 1, 328 pp., Book 2, 340 pp., Nedra, Moscow, 1990.

45. Scotese, C.R., *Paleogeographic Atlas, PALEOMAP Progress Report 90-0497*, 45 pp., Department of Geology, University of Texas at Arlington. Arlington, Texas, 1997.

46. Metelkin, D.V., I.V. Gordienko, and Xixi Zhao, Paleomagnetism of Early Cretaceous volcanic rocks from Transbaikalia: argument for Mesozoic strike-slip motions in Central Asian structure, *Geologiya i Geofizika (Russian Geology and Geophysics)*, **45**, 12, 1404–1417(1349–1363), 2004.

Editorial responsibility: V.A. Vernikovsky Received 13 September 2004